

Lincoln University College
Faculty of Computer Science and Multimedia

**PocketDRS: A Single-View 3D Trajectory
Reconstruction and Decision Review
System for Cricket**

A PROJECT PROPOSAL

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1 Introduction

Cricket has become one of the most data-driven and technology-assisted sports in the world. At the international level, technologies such as Hawk-Eye, UltraEdge, and ball-tracking systems are now deeply integrated into umpiring decisions and broadcast graphics. These systems help umpires review close calls related to Leg Before Wicket (LBW), caught behind decisions, and boundary checks. However, they require multiple high-speed cameras, specialized lenses, dedicated servers, and trained operators, which makes them financially and logistically out of reach for local clubs, schools, academies, and grassroots tournaments.

Most amateur and semi-professional matches still rely entirely on a single on-field umpire with no technological assistance. This increases the chances of wrong decisions, leads to disputes between players and officials, and can negatively affect the competitive fairness and enjoyment of the game. At the same time, the capabilities of consumer smartphones have improved significantly, with powerful CPUs and GPUs, built-in accelerators for machine learning, high-resolution cameras, and fast local networking features.

PocketDRS is a proposed mobile application that aims to bring core functionality of a Decision Review System onto a single smartphone, supplemented optionally by a second phone acting as an audio capture device. The central idea is to use the phone camera for video capture, computer vision for ball tracking and pitch calibration, and physics-based modelling to reconstruct and predict the ball trajectory in three dimensions. This reconstruction is then used to assist LBW decisions and edge detection, similar in spirit to professional DRS, but implemented in a low-cost and **fully offline** manner.

The project focuses on single-view 3D reconstruction, meaning that all tracking and prediction must be done from only one camera viewpoint. The pitch and stump geometry are used as strong constraints, and the system is designed to run entirely on-device without requiring internet access. This makes PocketDRS suitable for rural grounds and informal tournaments where stable connectivity cannot be guaranteed.

2 Problem Statement

Commercial DRS solutions solve the ball tracking problem using multiple synchronized cameras positioned around the ground. These systems triangulate the ball's 3D position from different views and feed the result into a physics engine to predict the ball's future path. This approach works very well but is extremely expensive, and it cannot be scaled down to a typical local ground.

Bringing DRS-style functionality onto a single smartphone introduces three main technical challenges:

- **Monocular Depth Ambiguity:** With only one camera, depth information is not

directly available. A point in the image could correspond to infinitely many points along a ray in 3D space. Recovering a realistic 3D trajectory from a single view requires using strong prior knowledge and constraints, such as the known dimensions of the cricket pitch and the assumption that the ball moves within a vertical plane for a given delivery angle.

- **Spin Estimation without High-Speed Cameras:** Professional systems often use high frame rate cameras or specialized sensors to measure the rotation of the ball. A typical smartphone camera running at 30 or 60 frames per second cannot capture the seam rotation of a fast delivery cleanly. Therefore, spin cannot be measured directly and must be inferred from the way the ball’s path curves in space due to aerodynamic forces like the Magnus effect.
- **Audio and Video Synchronization on Local Networks:** For caught behind decisions, it is critical to correlate a spike in the audio waveform (from a stump microphone) with the exact frame where the ball passes the bat. Over a Wi-Fi hotspot connection between two phones, network latency and jitter cause timing differences. The system must implement a synchronization mechanism that achieves frame-level precision without relying on internet network time servers.

The problem this project aims to solve is therefore: Can a single smartphone, possibly supported by a second phone as a stump microphone, reconstruct a cricket ball’s 3D trajectory accurately enough to assist LBW and edge decisions, using only on-device computation and **offline** communication between devices?

3 Objectives

The main objective is to design and implement a prototype Decision Review System that operates entirely on mobile devices and provides understandable, visual, and numerical support for cricket umpiring decisions.

3.1 General Objective

To develop an **offline, single-view, mobile-based Decision Review System** for cricket that can reconstruct and predict a ball’s trajectory, assist LBW decisions, and support edge detection using synchronized audio and video.

3.2 Specific Objectives

- To study existing professional DRS technologies, current research on monocular 3D reconstruction, and mobile-based tracking systems, and to identify key techniques that can be adapted to a smartphone environment.

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- To design and implement an **AR-based pitch calibration module** that uses known pitch and stump dimensions to estimate the camera pose using a **Perspective-n-Point (PnP) approach**.
 - To develop a ball detection and tracking module that can identify the ball in each video frame and produce a smoothed 2D trajectory using techniques such as background subtraction and Kalman filtering.
 - To create a **3D trajectory reconstruction and prediction module** that lifts the 2D trajectory into 3D space using the calibration information and fits a physics-based model incorporating gravity, drag, and the Magnus effect to estimate swing and spin.
 - To implement an **audio-video synchronization module** that uses time exchange messages between two phones on a local Wi-Fi connection to achieve frame-accurate alignment between stump microphone audio and camera video.
 - To design a user interface that visualizes the reconstructed and predicted trajectory, highlights pitching, impact, and wicket regions, and presents edge detection information in a format that umpires and players can easily interpret.

4 Methodology

The methodology covers requirement identification, feasibility analysis, system design, and the core algorithmic components. The development will follow an iterative approach: start with simple, working versions of each component and gradually refine them based on testing.

4.1 Requirement Identification

4.1.1 Study of Existing System

Existing professional systems such as Hawk-Eye typically deploy multiple synchronized cameras around the ground, all calibrated to a common 3D coordinate system. These systems perform stereo triangulation to compute the ball's 3D position over time and then pass the result into a physics engine. For this project, such hardware and infrastructure are not available. This project replaces the multi-camera architecture with a single camera and replaces the large server-based processing pipeline with optimized **on-device algorithms**. Instead of relying on triangulation between cameras, the system uses camera calibration, geometric constraints, and motion models to reconstruct the ball trajectory from a single viewpoint.

4.1.2 Literature Review

The literature related to this project can be grouped into the following themes:

- **Camera Calibration and PnP:** The **Perspective-n-Point (PnP)** problem is highly relevant, as a cricket pitch provides known 3D points (stumps, creases) whose corresponding 2D image projections can be used to estimate camera pose.
- **Monocular 3D Reconstruction and Object Tracking:** Research confirms the feasibility of reconstructing trajectories using single-camera video when strong constraints (fixed field dimensions and ballistic motion) are available.
- **Ball Physics and the Magnus Effect:** The trajectory is influenced by gravity, aerodynamic drag, and the **Magnus effect** (spin-induced force). Physics models allow approximation of the spin vector ($\vec{\omega}$) from the observed curvature, bypassing the need for high-speed cameras.
- **Time Synchronization over Local Networks:** Variants of the Network Time Protocol (NTP) over local UDP communication can be adapted to achieve a common clock between two devices on a **Wi-Fi hotspot** with sub-millisecond precision, essential for audio-video alignment.

4.1.3 Requirement Analysis

Functional Requirements: The system shall capture video, perform AR-based calibration, track the ball, reconstruct its 3D path, predict LBW outcome, record and synchronize audio for edge detection, and present decision information visually.

Non-Functional Requirements: The system must work **fully offline**, processing time should ideally be below 5 seconds, reconstructed trajectory accuracy must be sufficient for practical LBW judgment, and the user interface must be simple and intuitive.

4.2 Feasibility Study

4.2.1 Technical Feasibility

The system is technically feasible using the following stack:

- **Platform/UI:** Flutter for cross-platform mobile development.
- **Computer Vision:** Integration of OpenCV through native wrappers for PnP, tracking, and filtering.
- **Physics:** Lightweight implementation of the ball dynamics equation using mobile CPU resources.

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- **Machine Learning:** The use of optimized models like **TensorFlow Lite (TFLite)** for ball detection ensures high performance and **fully offline** computation.

4.2.2 Operational Feasibility

The solution fits the amateur environment:

- **Setup:** A single phone mounted on a tripod, easily calibrated using AR markers against the pitch lines.
- **Usability:** Simple "record" or "review" button presses allow umpires with basic smartphone familiarity to operate the system.
- **Audio:** The optional second phone (stump mic) connects automatically via local hotspot, minimizing setup complexity.

4.2.3 Economic Feasibility

The system is economically highly feasible as it requires ****no specialized sensors or cameras****. Development relies on freely available open-source software (OpenCV, TFLite, Flutter), resulting in zero licensing costs. The core application can be widely reused, making the cost per match negligible.

4.3 High Level Design of System

The system is divided into four main layers: calibration, tracking, physics, and presentation.

Figure 4.3: PocketDRS data flow from video input through calibration, tracking, physics analysis, and final decision visualization.

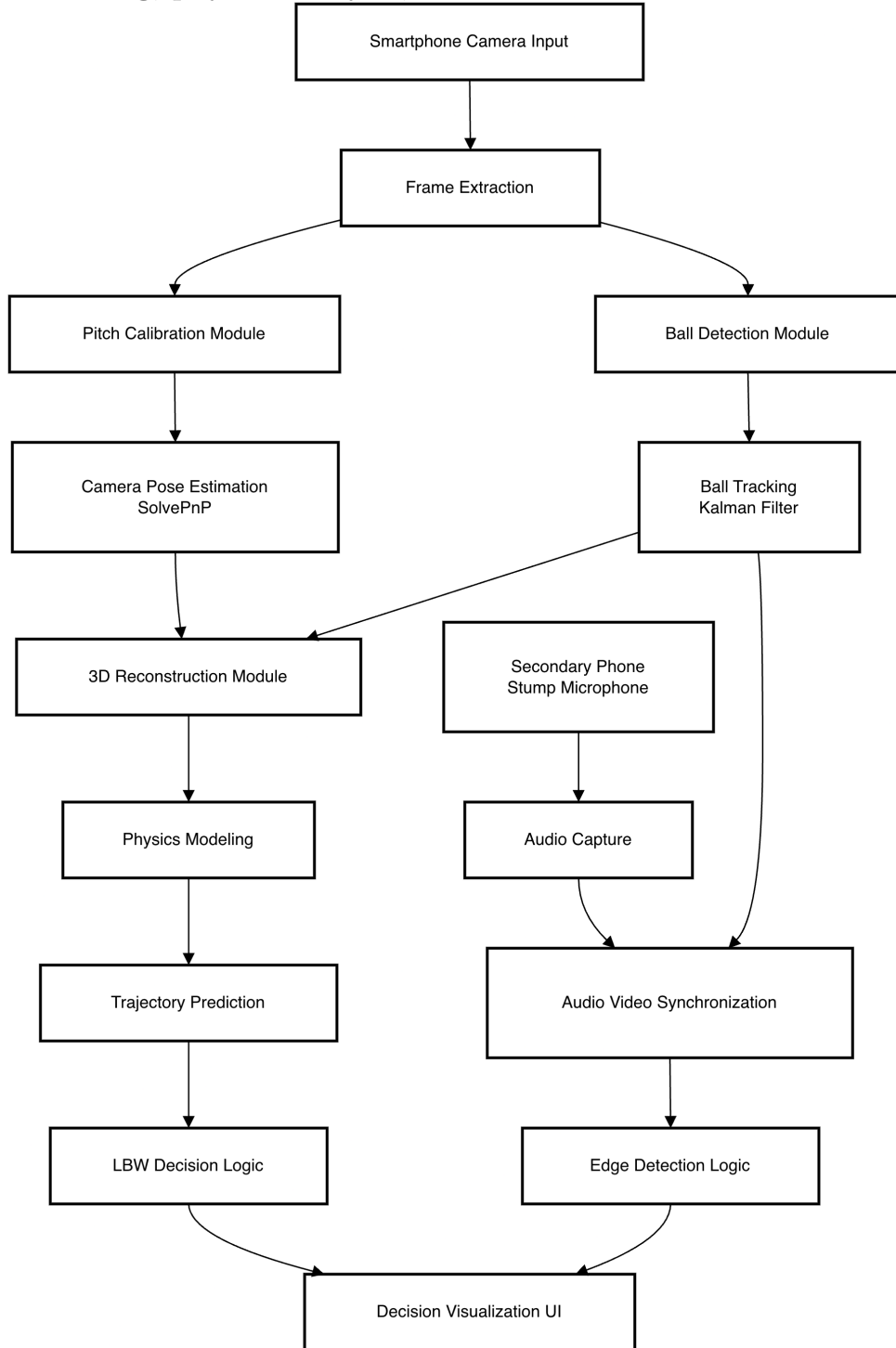


Figure 1: Layered architecture showing the path from video input to final decision data output.

The **Physics Prediction Layer** uses the tracked 3D velocity and spin data to solve the vector equation of motion, accounting for gravity, air resistance (drag), and the crucial

Magnus effect:

$$m\vec{a} = m\vec{g} - C_D|\vec{v}|^2\hat{v} + C_M(\vec{\omega} \times \vec{v})$$

Here, m is mass, $\vec{a}$ is acceleration, $\vec{g}$ is gravity, C_D is the drag coefficient, and C_M is the Magnus coefficient related to the spin vector $\vec{\omega}$. This calculation enables accurate prediction of swing and spin movement.

5 Gantt Chart

The development will be carried out over approximately four months.

Activity	Dec '25	Jan '26	Feb '26	Mar '26
Literature Review and Detailed Analysis	X			
Pitch Calibration Module (PnP, Camera Pose)		X		
Ball Detection and Tracking Module		X	X	
3D Reconstruction and Physics Modelling			X	
Audio Synchronization Module and UI Integration				X
Field Testing, Refinement, and Final Documentation				X

6 Expected Outcome

Upon completion, the project is expected to deliver the following outcomes:

- **WORKING PROTOTYPE APPLICATION:** An Android application (APK) that can record or load a video of a delivery, perform calibration, track the ball, reconstruct its trajectory, and visualize the result.
- **LBW ASSISTANCE:** For suitable camera angles, the system will show whether the ball pitched in line, whether it struck the batsman in line, and whether the predicted trajectory would hit the stumps, providing a clear LBW decision suggestion.
- **EDGE DETECTION SUPPORT:** When used with a second device as a stump microphone, the system will synchronize audio and video to identify sounds that

coincide precisely with the ball passing the bat, helping umpires judge possible edges.

- **DOCUMENTATION AND EVALUATION:** A detailed report documenting the algorithms, design decisions, implementation details, test results, and limitations.
- **FOUNDATION FOR FUTURE WORK:** The project will serve as a base for enhancements such as deep learning based ball detectors or automatic bat and pad detection.

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